

# VARUN NAUDIYAL

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## PROFILE

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I am a Master's student who works in the field of **Multi-Messenger Astrophysics**. My interests lie in using **machine learning and numerical simulations to explore the physics of Neutron Stars** and their evolution in various environments. Alongside this, I am also interested in understanding the **evolution and formation of stellar binary systems** and how their interactions can lead to the formation of compact binaries, whose mergers we ultimately observe through gravitational waves and electromagnetic counterparts such as kilonovae.

## PROJECTS

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**Physics-Informed Neural Networks for Neutron Star EOS Inference (Ongoing)** Aug 2026 – Present  
*Supervised by Dr. Ritam Mallick, IISER-B* *Master's Thesis*

**Problem Statement:** Inferring the dense-matter equation of state (EOS) from astrophysical observations is an inverse problem. Bezerra et al. tackle it with coupled physics-informed neural networks (PINNs), training an EOS network jointly with a structure network that enforces the TOV equations as a differential constraint, so the EOS is learned directly from mass–radius data. This thesis extends that framework by:

- **Tidal deformability:** adding it alongside the TOV equations as a second constraint, since gravitational-wave events such as GW170817 constrain it independently of mass and radius, and requiring the inferred EOS to be consistent with both;
- **Bayesian inference:** giving the inferred EOS calibrated uncertainty, in place of the seed-to-seed spread used to assess reproducibility.
- **First-order phase transitions:** developing a framework for the EOS discontinuities they introduce, which smooth networks struggle to represent;

**Rotational Energy Budget of Supramassive Neutron Star Merger Remnants** Jun 2026 – Jul 2026  
*Supervised by Dr. Ritam Mallick, IISER-B* *Summer Project*

- A long-lived supramassive remnant can spin down and inject rotational energy into the ejecta, “super-charging” the kilonova. Whether GW170817’s remnant could have supplied the observed energy budget constrains how close it must have been to the maximum mass, and hence the EOS.
- Built a Python pipeline around the RNS code that takes a tabulated EOS and computes the static, mass-shedding and constant-baryonic-mass spin-down sequences, from the Keplerian limit to the turning point. This includes the extractable rotational energy for any EOS.
- Tested whether a long-lived supramassive remnant is compatible with the energy budget observed in GW170817, finding that the extractable spin-down energy is small enough only for configurations near the maximum mass, which reproduces the published constraint.

**Building Neutron Star Equations of State from Relativistic Mean-Field Theory** Aug 2026 - Present

- Studied how nuclear microphysics (meson–baryon couplings, particle content, beta equilibrium) determines the stiffness of the EOS and, through it, the neutron star maximum mass and compactness.
- Developing a self-consistent relativistic mean-field solver for beta-equilibrated matter with the baryon octet coupled to  $\sigma$ ,  $\omega$ ,  $\rho$  mesons, with electrons and muons enforcing charge neutrality.
- Extending the framework to hyperon-rich and hybrid (hadron-quark) matter to compare their effect on the maximum mass.

## Exploring Neutron Stars in $R^2$ Gravity

Jun 2025 – Jul 2025

*Supervised by Mr. Asim Kumar Saha and Dr. Ritam Mallick, IISER-B*

*Summer Project*

- Investigated whether Starobinsky  $R^2$  gravity admits well-behaved neutron star solutions, and how the scalaron modifies the mass–radius relation relative to GR.
- Derived the  $R^2$  field equations from the action (by hand, cross-checked with a generalised Mathematica pipeline), extended the TOV equations to  $f(R)$  gravity, and solved them with a shooting method for a polytropic EOS.
- Showed that exterior solutions are exponentially sensitive to the central Ricci scalar, so matching to a physical vacuum requires fine-tuning of the shooting parameter (the scalaron fine-tuning problem).

## Numerical Relativity in 1 Dimension

Aug 2025 – Nov 2025

*Supervised by Mr. Hrishikesh Borthakur and Dr. Ritam Mallick, IISER-B*

*Semester Project*

- Studied 3+1 (ADM) numerical relativity and derived Oppenheimer–Snyder collapse in pressureless-dust, radiation-pressure and slow-rotation approximations.
- Evolved a spherical polytrope ( $K = 100$ ,  $\Gamma = 2$ ) with GR1D; verified stability by matching central-density oscillation modes to theory.

## TECHNICAL SKILLS

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|--------------------------|----------------------------------------------------------------------------|
| <b>Languages</b>         | Python, Fortran90, Mathematica, L <sup>A</sup> T <sub>E</sub> X            |
| <b>Python Libraries</b>  | NumPy, SciPy, Matplotlib, Pandas, PyTorch                                  |
| <b>Codes</b>             | RNS (Rotating Neutron Stars), GR1D (GR 1D Hydrodynamics)                   |
| <b>Numerical Methods</b> | ODE integration (shooting, Runge–Kutta), root-finding, finite differencing |

## RELEVANT COURSEWORK

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|                                       |                                         |
|---------------------------------------|-----------------------------------------|
| (PHY616) General Relativity           | (PHY630) Cosmology I                    |
| (PHY633) Introduction to Astrophysics | (PHY634) Advanced Statistical Mechanics |
| (PHY615) Quantum Field Theory I*      |                                         |

\*ongoing

## EXTRACURRICULARS

### Core Committee Member

Aug 2024 – Mar 2025

*IISER-Bhopal Astronomy Club (IBAC)*

- Contributed to the annual summer magazine, gave a public lecture on star formation, and ran public telescope sessions with Dobsonian reflectors.

### Core Committee Member, Design & Content

Aug 2024 – Nov 2024

*Armageddon — Annual Tech-Fest, IISER Bhopal*

- Led the Design and Content team: posters, banners, the official fest logo, and scripts for event hosts.

### Peer Counselor

Aug 2023 – Mar 2024

*IISER-Bhopal Student Wellness & Counselling Team*

- Provided confidential peer support to students and helped organise outreach to reduce stigma around mental health.